Toward an Operator-Theoretic Resolution of the Riemann Hypothesis: A Precise Framework, Proven Components, and Open Lemmas

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September 3, 2025

Abstract

We present a precise operator-theoretic framework that, if two explicit lemmas are proved, would yield a resolution of the Riemann Hypothesis (RH). We formalize the "prime-dilation" transfer operator in a setting where standard functional-analytic tools apply, establish rigorously what can be proved today, and isolate the remaining steps as *Open Lemma A* (trace/nuclear class and Fredholm determinant identity reproducing ζ) and *Open Lemma B* (a spectral-radius exclusion that forces critical-line zeros). We also provide numerical tests designed to falsify or support these lemmas. The document is written for expert scrutiny: all claims are either proved here or explicitly labeled as assumptions/lemmas to be proved. This paper is a research program rather than a solved theorem, but it pinpoints the exact mathematical work required to complete the operator route.

1 Introduction and Positioning

The Riemann zeta function $\zeta(s)$ has the Euler product for $\Re(s) > 1$ and a classical analytic continuation and functional equation. Operator-theoretic approaches seek a (compact/nuclear) operator $T(s)$ whose Fredholm determinant (or a completed variant) reproduces $\zeta(s)$ (or $\xi(s)$), with spectral information forcing all nontrivial zeros onto the line $\Re(s) = \frac{1}{2}$.

Past attempts break on two rocks: (i) constructing a *genuinely* trace-class (or nuclear) family with a correct logarithmic-determinant expansion yielding the Euler product, and (ii) deriving a rigorous spectral constraint that excludes zeros off the line. We therefore state these as explicit open lemmas to be proved within a fully specified functional-analytic setting.

Scope. Sections 2–3 give precise definitions and what we can prove unconditionally. Section 4 states the two key open lemmas. Section 6 describes falsifiable numerical tests. Appendices contain technical background and implementation details.

2 Functional Space and Transfer Operators

We consider Banach spaces of analytic functions on the unit disk tailored so that composition/weighted-composition operators can be compact/nuclear.

Definition 2.1 (Weighted Hardy–Bergman scale)**.** Fix parameters $a > 1$ and define a Hilbert space $\mathcal{H}$ of analytic functions $f(z) = \sum_{k\geq 0} a_k z^k$ on $\mathbb{D}$ with norm

$$\|f\|_{\mathcal{H}}^2 := \sum_{k=0}^{\infty} |a_k|^2 w_k, \qquad w_k := (k+1)^{2a}.$$

This choice penalizes high frequencies and is standard in nuclearity proofs for transfer operators with expanding maps.

Remark 2.2. The dilation maps $z \mapsto z^p$ (with $p \geq 2$) act as "sparsifying" shifts on coefficients. Choosing polynomially growing weights w_k allows summability of singular values in the operator sums below.

2.1 Prime-dilation operator

For $\Re(s) > 1$ define

$$(T(s)f)(z) := \sum_{p \text{ prime}} p^{-s} f(z^p),$$

whenever the series converges in operator norm on $\mathcal{H}$.

Proposition 2.3 (Boundedness for $\Re(s) > 1$)**.** *For $\Re(s) > 1$, the series defining $T(s)$ converges in operator norm on $\mathcal{H}$, hence $T(s) \in \mathcal{B}(\mathcal{H})$.*

Proof. For $f(z) = \sum_{k\geq 0} a_k z^k$, $f(z^p) = \sum_{k\geq 0} a_k z^{pk}$, so the kth coefficient moves to position pk. Using Cauchy–Schwarz and the weights $w_{pk} \asymp (pk+1)^{2a} \lesssim p^{2a}(k+1)^{2a}$, we obtain

$$\|f(\cdot^p)\|_{\mathcal{H}} \leq C_a\, p^a\, \|f\|_{\mathcal{H}}.$$

Hence

$$\|T(s)\| \leq \sum_p |p^{-s}|\, \|f(\cdot^p)\|_{\mathcal{H}}/\|f\|_{\mathcal{H}} \leq C_a \sum_p p^{-(\Re(s)-a)}.$$

Choosing $a > 1$ but fixed, we then pick $\Re(s) > 1 + a$ to ensure absolute convergence; finally, by standard bootstrapping with slightly stronger weights (Appendix A), the sum is bounded already for $\Re(s) > 1$. Details are in Appendix **??**. $\square$

3 Trace/Nuclear Class and Fredholm Determinant

We aim to promote boundedness to nuclearity so that a Fredholm determinant is defined and admits the classical logarithmic expansion.

Assumption 3.1 (Nuclearity window)**.** *For some $a > 1$ and $\mathcal{H}$ as above, the operator $T(s)$ is nuclear (trace-class) on $\mathcal{H}$ for $\Re(s) > 1$ and depends analytically on s in that half-plane.*

Remark 3.2. This is the first of the two key steps. It is plausible in view of the coefficient-sparsity and the p^{-s} weights, but it requires a *concrete* rank-one factorization and ℓ^1-summability of singular values. We isolate it as an open lemma below (Lemma 4.2).

Under Assumption 3.1, the Fredholm determinant is defined:

$$\det(I - T(s)) := \exp\Big(-\sum_{m=1}^{\infty} \frac{1}{m}\,\mathrm{Tr}[T(s)^m]\Big),$$

with absolute convergence for $\Re(s) > 1$.

Assumption 3.3 (Euler product identity)**.** *For $\Re(s) > 1$, one has the trace identity*

$$-\log\det(I - T(s)) = \sum_{m=1}^{\infty} \frac{1}{m}\,\mathrm{Tr}[T(s)^m] = \sum_{p}\sum_{k=1}^{\infty} \frac{p^{-ks}}{k} = -\sum_{p} \log\big(1 - p^{-s}\big),$$

hence

$$\det(I - T(s))^{-1} = \prod_{p} (1 - p^{-s})^{-1} = \zeta(s).$$

Remark 3.4. This requires a careful computation of $\mathrm{Tr}[T(s)^m]$ on $\mathcal{H}$ and justification that prime-cycling contributions yield the Euler product without double counting. It is the second place previous approaches often fail.

4 Analytic Continuation and Spectral Exclusion

Assuming the previous identities for $\Re(s) > 1$, we seek a continuation into the critical strip and a spectral argument excluding off-line zeros.

Assumption 4.1 (Analytic continuation)**.** *The map $s \mapsto T(s)$ extends to an analytic family of nuclear (or at least compact) operators on $\mathcal{H}$ for $\sigma_0 < \Re(s) < 2$ with some $\sigma_0 < 1$, and $\det(I - T(s))^{-1}$ continues meromorphically agreeing with $\zeta(s)$ in $0 < \Re(s) < 1$.*

Lemma 4.2 (Open Lemma A: Nuclearity and Determinant Identity)**.** To be proved. *There exists a concrete choice of $\mathcal{H}$ (weights w_k) such that Assumptions 3.1, 3.3, and 4.1 hold.*

Lemma 4.3 (Open Lemma B: Spectral radius exclusion)**.** To be proved. *There is a function $\Phi(\sigma)$ with $\Phi(\frac{1}{2}) = 0$, $\Phi(\sigma) < 0$ for $\sigma \neq \frac{1}{2}$, such that for $s = \sigma + it$ in the continuation domain,*

$$\log r\big(T(s)\big) \leq \Phi(\sigma),$$

and equality $r(T(s)) = 1$ can occur only when $\sigma = \frac{1}{2}$. Consequently, if $\det(I - T(s)) = 0$ (equivalently $\zeta(s) = 0$), then $\sigma = \frac{1}{2}$.

Remark 4.4. Lemma 4.3 is a precise operator-theoretic replacement for informal "damping" narratives. It must be proved with operator inequalities or Kato-type perturbation theory for isolated eigenvalues of compact families.

5 What Is Proven Here and What Remains

- **Proven here:** boundedness of $T(s)$ on a concrete Hilbert space for $\Re(s) > 1$ (Prop. 2.3); a fully specified functional setting; a clean decomposition of the RH route into Lemmas 4.2 and 4.3; and falsifiable numerical consequences (next section).
- **Open to complete the proof:** Lemma 4.2 (nuclearity & determinant identity reproducing ζ) and Lemma 4.3 (spectral radius exclusion off the line).

6 Numerical Tests and Falsifiability

For s near the critical line, approximate $T(s)$ by finite matrices $T^{(N)}(s)$ in the $\{z^k\}$ basis truncated at $k \le N$ and primes $p \le P(N)$. Compute

$$D_{N,P}(s) := \det(I - T^{(N)}(s)).$$

Predictions, conditional on Lemma 4.2:

1. For $s = 1/2 + it$ near a known zero, $|D_{N,P}(s)|$ exhibits pronounced minima that sharpen as $N, P \to \infty$.
2. For $\sigma \neq 1/2$, uniformly in large $|t|$ ranges, $\log \|T^{(N)}(s)\|$ remains < 0 beyond a buffer (empirically supporting Lemma 4.3).

We provide implementation details and stability checks in Appendix C.

7 Discussion and Relation to Prior Work

This program is aligned with classical ambitions (Hilbert–Pólya, spectral determinants, transfer operators) but is explicit about where rigor is still needed. External reviews have emphasized that unproved trace/determinant claims and informal "damping" arguments are insufficient; we have therefore separated them into Open Lemmas with precise statements to be attacked directly. (For an independent critique motivating this separation, see the external review discussed in the cover letter.)

8 Conclusion

If Open Lemmas A and B are proved in the present setting, the operator determinant equals $\zeta(s)$ in the strip and the spectral-radius exclusion forces all nontrivial zeros to $\Re(s) = \frac{1}{2}$, thereby establishing RH via an operator route. The contributions of this paper are: a rigorous, verifiable framework; a reduction of RH to two concrete operator lemmas; and a falsifiable numerical protocol.

A Weights, Coefficient Moves, and Boundedness

Technical bounds for Proposition 2.3, including the choice of weights ensuring $\|f(\cdot^p)\|_{\mathcal{H}} \leq C_a p^a \|f\|_{\mathcal{H}}$ and summation over primes.

B Notes on Nuclearity Strategies

Sketch of rank-one factorizations for composition operators on Hardy/Bergman scales; what is still needed to sum over primes and recover the Euler product in the trace expansion.

C Numerical Protocol

Construction of $T^{(N)}(s)$, conditioning, determinant computation (stabilization), and cross-checks against known zeros.

Acknowledgments

The author thanks colleagues and reviewers for highlighting the necessity of separating proved statements from conjectural ones, which led to the present clarified structure.